# Supplementary Material

Certified Soft-Output List Evaluation for Coded SP-AFDM With Fractional Doppler

20 September 2026

This supplement collects physical/statistical identities that support the paper without constituting additional proposed receiver methods. It also records the numerical contracts, implementation qualifications, and evidence provenance. The main paper's only proposed method is certified computation-adaptive evaluation of a fixed codeword list. Section numbers, notation, and signs here are consistent with that method; no historical support-repair receiver is carried forward.

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# S1 Physical convention and exact finite representation

## S1.1 Positive chirps, positive Doppler, and the CPP

The paper uses the negative-exponent unitary DFT and positive chirps:

$$F_{mk} = N^{-1/2} e^{-j2\pi mk/N}, \qquad A = \Lambda_{c_2} F \Lambda_{c_1}, \quad \Lambda_c = \mathrm{diag}(e^{j2\pi c n^2}). \tag{S1}$$

The physical path is $U(l,\nu) = AC_l\Delta_\nu\Pi_l A^{\mathsf{H}}$, with positive Doppler exponential and the CPP in main (4). These operators are unitary individually. The use of $A^{\mathsf{H}}$ at both the transmitted-vector and received-vector conversion boundaries preserves energy and proper white noise. The AFDM construction and chirp-dependent input–output structure originate in [1]; fixing signs here prevents mixing two equivalent but numerically different parameter conventions.

Let

$$D_N(t) = \frac{1}{N}\sum_{n=0}^{N-1} e^{j2\pi nt/N}. \tag{S2}$$

Direct expansion gives

$$[U(l,\nu)]_{mk} = e^{j2\pi[-c_1 l^2 - kl/N + c_2(m^2-k^2)]} D_N(k-m+\nu+2Nc_1 l). \tag{S3}$$

To verify the phase, insert $A$ and $A^{\mathsf{H}}$ on the two sides of the delayed time-domain action. On nonwrapped samples, $c_1[n^2-(n-l)^2] = 2c_1 nl - c_1 l^2$. The Fourier phases supply $(k-m)n/N - kl/N$, and Doppler supplies $\nu n/N$. On wrapped samples the CPP cancels the extra chirp phase associated with replacing $n-l$ by $(n-l)$ mod $N$. Summing over $n$ yields (S3).

The original foundation used negative chirps and negative Doppler exponentials. Its formulas translate by $(c_1, c_2, \nu) \mapsto (-c_1, -c_2, -\nu)$, not by changing only the Doppler sign. For integer $\alpha = 2Nc_1$ and integer $\nu = q$, (S3) is nonzero only when

$$k = m - q - \alpha l \pmod N. \tag{S4}$$

Thus $[U(l,q)x]_m = u_{lq}[m]x_{m-s}$ with $s = q + \alpha l \pmod N$ and $|u_{lq}[m]| = 1$. Distinct physical coordinates need not have distinct effective shifts. This fact matters for the CE special case below, but is not a prerequisite for the proposed time-domain evaluator.

## S1.2 Fractional paths have an exact integer expansion

For every real $\nu$, define

$$c_q(\nu) = \frac{1}{N}\sum_{n=0}^{N-1} e^{j2\pi(\nu-q)n/N}. \tag{S5}$$

Finite Fourier inversion and Parseval give

$$U(l,\nu) = \sum_{q=0}^{N-1} c_q(\nu) U(l,q), \qquad \sum_q |c_q(\nu)|^2 = 1. \tag{S6}$$

For integer Doppler, one coefficient remains. For fractional Doppler, the expansion is generally dense. Hence continuous parameter estimation is not required for exact representation of a sampled path, but a small physical path count does not imply a comparably small integer coefficient count.

For pairs $(l,q)$ over the complete $N \times N$ grid,

$$\mathrm{tr}\{U(l,q)^{\mathsf{H}} U(l',q')\} = N\,\mathbf{1}_{l=l'}\mathbf{1}_{q=q'}. \tag{S7}$$

Unitary conjugation preserves trace. Unequal delays give a nonzero cyclic shift with zero trace; equal delays cancel the CPP, leaving an integer-exponential sum. Physical contributions sharing the same integer coordinate must be merged first:

$$a_{lq} = \sum_{r:l_r=l} h_r c_q(\nu_r), \qquad H = \sum_{l,q} a_{lq} U(l,q). \tag{S8}$$

The Frobenius-orthogonal projection onto an index set $\mathcal{S}$ then obeys

$$\left\| H - H_{\mathcal{S}}^{F} \right\|_F^2 = N \sum_{(l,q)\notin\mathcal{S}} |a_{lq}|^2. \tag{S9}$$

Separate path-tail energies cannot generally be added because their coherent cross terms can survive. Renormalizing retained coefficients does not restore deleted phases or couplings.

The proposed evaluator never uses (S9) as a pruning rule. It evaluates complete fractional path responses and prunes *candidate computations*, not physical channel components. This distinction separates its approximation contract from truncating the propagation model to match a sparse receiver.

# S2 SP nuisance statistics and why no data-Gaussian approximation is required

## S2.1 What superimposition prevents

If allowed data vectors span $\mathbb{C}^N$, any fixed linear map satisfying $LHd = 0$ for every allowed $d$ also satisfies $LHp = 0$. This follows because $LH$ vanishes on the whole input space. Exact universal linear removal of unknown data would remove the pilot response too. The result is not an impossibility theorem for nonlinear joint inference, restricted-data designs, or independent training observations.

For one fixed gain, independent zero-mean unit-variance $z_n$, and

$$y_n = h(\sqrt{P_p} + \sqrt{P_d} z_n) + w_n, \qquad \widehat{h} = \frac{1}{N\sqrt{P_p}} \sum_n y_n, \tag{S10}$$

the estimator has conditional MSE $(P_d|h|^2 + \sigma^2)/(NP_p)$. Its residual data term is an estimator-specific floor, not a universal SP bound. Data-aided iterative estimators use information that this averaging estimator discards [2, 3].

## S2.2 Conditional covariance is not an ensemble covariance

For zero-mean data with covariance $V_0$, independently of thermal noise,

$$\mathrm{Cov}(Hd + w \mid H) = HV_0H^{\mathsf{H}} + \sigma^2 I. \tag{S11}$$

This identity does not assume Gaussian data. Static two-tap $H = h_0 I + h_1 \Pi$ gives shifted cross terms in $HH^{\mathsf{H}}$ and usually colored interference. Conversely, a single fractional path $H = hU$ with unitary $U$ gives white covariance for $V_0 = P_d I$. Fractional Doppler is neither necessary nor sufficient for interference color.

For independent, zero-mean path gains and isotropic data, averaging over those gains removes cross-path terms and can make (S11) proportional to $I$. This different conditioning does not make every realized channel's interference white. If the code introduces symbol correlations, its actual covariance must replace $P_d I$.

### S2.3 Truncation creates dependent residual terms

Writing $H = H_R + H_T$ gives the omitted response $H_T p + H_T d$. The first term is coherent pilot-model error. For isotropic data,

$$\mathrm{Cov}(H_R d, H_T d) = P_d H_R H_T^{\mathsf{H}}. \tag{S12}$$

Frobenius orthogonality $\mathrm{tr}(H_R^{\mathsf{H}} H_T) = 0$ does not force this matrix to vanish. Replacing the omitted response by independent white noise is therefore a receiver approximation requiring validation, not an identity. No such replacement is made in the main algorithm.

### S2.4 A large block need not Gaussianize fractional-path interference

For independent proper QPSK symbols with energy $P_d$ and $z = \sum_n a_n d_n$,

$$\mathbb{E}|z|^4 - 2(\mathbb{E}|z|^2)^2 = -P_d^2 \sum_n |a_n|^4. \tag{S13}$$

Independence leaves the paired fourth-moment terms; properness removes $\mathbb{E}d_n^2$, and constant modulus gives $\mathbb{E}|d_n|^4 = P_d^2$ instead of the Gaussian $2P_d^2$. For $\nu = k + \delta$, the finite expansion satisfies

$$\sum_q |c_q(\nu)|^4 = 1 - \frac{2}{3}\left(1 - \frac{1}{N^2}\right)\sin^2(\pi\delta). \tag{S14}$$

One proof uses the normalized circular autocorrelation of $e^{j2\pi\nu n/N}$: its squared magnitude at lag $r$ is $1 - 4\sin^2(\pi\nu) r(N-r)/N^2$. Apply Parseval and sum $\sum_{r=0}^{N-1} r(N-r) = N(N^2-1)/6$. At half-bin Doppler and large $N$, the effective number of mixing terms is only three, and the normalized cumulant in (S13) approaches $-1/3$.

This qualification is retained to prevent a false justification of the proposed likelihood. Conditional on a complete codeword, the transmitted data are fixed. Its Gaussian marginal likelihood follows from the declared Gaussian gain prior and Gaussian thermal noise, *not* from claiming that unknown coded QPSK interference becomes Gaussian as $N$ grows.

## S3 How posterior data statistics enter a conventional CE update

This section explains why soft information is a material receiver output. It is not an additional proposed CE algorithm and must not be substituted for the posterior gain mean used in the main paper.

For any frozen working distribution $q(d)$, write

$$\mu_q = \mathbb{E}_q d, \qquad V_q = \mathbb{E}_q[(d-\mu_q)(d-\mu_q)^{\mathsf{H}}], \tag{S15}$$

and let $H(g) = \sum_{i=1}^{S} g_i U_i$ and $\Phi_\mu = [U_1(p+\mu_q), \ldots, U_S(p+\mu_q)]$.

**Proposition S1** (Complete-data quadratic)**.** *For fixed observations and operators,*

$$\mathbb{E}_q \left\| y - H(g)(p+d) \right\|^2 = \left\| y - \Phi_\mu g \right\|^2 + g^{\mathsf{H}} \Gamma_q g, \qquad (\Gamma_q)_{ij} = \mathrm{tr}(U_i^{\mathsf{H}} U_j V_q), \tag{S16}$$

*and* $\Gamma_q \succeq 0$*. Adding* $\lambda \left\| g - g_0 \right\|^2$ *gives*

$$(\Phi_\mu^{\mathsf{H}} \Phi_\mu + \Gamma_q + \lambda I) g_q = \Phi_\mu^{\mathsf{H}} y + \lambda g_0. \tag{S17}$$

*Proof.* Substitute $d = \mu_q + e$ and expand the squared norm. The cross term vanishes because $\mathbb{E}_q e = 0$. The uncertainty term is $\mathrm{tr}[H(g) V_q H(g)^{\mathsf{H}}] = g^{\mathsf{H}} \Gamma_q g \geq 0$. Differentiating with respect to $g^*$ gives (S17). Uniqueness requires the normal matrix to be positive definite. □

No Gaussian, symbol-independence, or properness assumption on $q$ is needed. Its pseudo-covariance does not independently enter this proper complex squared norm. The uncertainty penalty is zero-centered; increasing a proximal penalty around $g_0$ changes both sides of (S17) and is not equivalent to adding $\Gamma_q$.

If $[U_i x]_n = u_i[n] x_{n-s_i}$, $|u_i[n]| = 1$, and the effective shifts $s_i$ are distinct modulo $N$, then diagonal $V_q = \mathrm{diag}(v)$ implies

$$\Gamma_q = \Big( \sum_n v_n \Big) I_S. \tag{S18}$$

Indeed,

$$(\Gamma_q)_{ij} = \sum_n u_i[n]^* u_j[n] (V_q)_{n-s_j, n-s_i}. \tag{S19}$$

Distinct shifts eliminate off-diagonal contributions when $V_q$ is diagonal. A fractional physical-path operator is not generally monomial in the DAFT domain and must not inherit (S18) without those conditions.

A normalized singleton list has zero data covariance even if its word is the complete posterior's maximizer. It can therefore preserve MAP while changing (S17). Conversely, the main paper's gain-mean bound concerns

$$\sum_d P_B(d \mid y)\, \mathbb{E}[g \mid d, y], \tag{S20}$$

which integrates gain uncertainty. It is not a theorem about every approximation to (S17); expectation and matrix inversion do not generally commute.

For weighted complete candidates, correlated $\Gamma_q$ can be formed without storing an $N \times N$ covariance. With $B_d = [U_1(d - \mu_q), \ldots, U_S(d - \mu_q)]$,

$$\Gamma_q = \sum_d w_d B_d^{\mathsf{H}} B_d. \tag{S21}$$

This is positive semidefinite by construction. Obtaining the words and their trustworthy weights remains part of the cost.

# S4 Iteration, noise normalization, and information reuse

## S4.1 Exact EM is a different conditional guarantee

For fixed channel coordinates and a prior $P_{\mathcal{C}}(d)$,

$$p(y, d \mid g) = P_{\mathcal{C}}(d) (\pi\sigma^2)^{-N} e^{-\|y - H(g)(p+d)\|^2/\sigma^2}. \tag{S22}$$

The free-energy identity is

$$\begin{aligned} &\sigma^{-2} \mathbb{E}_q \|y - H(g)(p+d)\|^2 + D_{\mathrm{KL}}(q \| P_{\mathcal{C}}) \\ &\quad = -\log p(y \mid g) + D_{\mathrm{KL}}(q \| p(d \mid y, g)) + \text{constant}. \end{aligned} \tag{S23}$$

When the E-step computes $q_t = p(d \mid y, g_t)$ exactly and the M-step decreases its complete-data loss, (S23) proves $p(y \mid g_{t+1}) \geq p(y \mid g_t)$, the classical EM result [4]. It does not imply global convergence, samplewise true-channel MSE improvement, or decreasing FER. Approximate decoder messages, changing supports, arbitrary damping, or omitted $\Gamma_q$ do not inherit this statement automatically.

The main algorithm evaluates a frozen list and frozen gain prior; it is not an EM loop. Replacing full-list soft output by a certified approximation controls one call's stated outputs. The next iteration can alter its model and list, so a multi-iteration error claim requires a separate propagation argument.

### S4.2 LLR units and damping

For $z = x + n$, $n \sim \mathcal{CN}(0, \sigma_n^2)$, uniform QPSK $x \in \{\pm a \pm ja\}$, and positive LLR indicating a positive component,

$$L_I = 4a\Re z/\sigma_n^2, \qquad L_Q = 4a\Im z/\sigma_n^2. \tag{S24}$$

Under independent bit probabilities, $t_I = \tanh(L_I/2)$ and $t_Q = \tanh(L_Q/2)$ give

$$\mu = a(t_I + jt_Q), \qquad v = a^2(2 - t_I^2 - t_Q^2). \tag{S25}$$

These moments already include data power. They cannot be rescaled by $P_d$ a second time. The main list probabilities instead preserve the complete codeword constraints and need not factorize.

If damping is interpreted as a mixture $q = (1-\gamma)q_0 + \gamma q_1$, its consistent covariance is

$$V = (1-\gamma)V_0 + \gamma V_1 + \gamma(1-\gamma)(\mu_0 - \mu_1)(\mu_0 - \mu_1)^{\mathsf{H}}. \tag{S26}$$

Damping only means while leaving an unrelated covariance does not describe this mixture. The certified evaluator does not introduce such damping internally.

For a decoder invocation, APP minus its actual input can define an extrinsic replacement at a documented interface. Replacing old information prevents explicit accumulation of that same input as fresh evidence; it does not make messages in a loopy graph statistically independent. In the present experiments, decoder reliabilities generate candidates and are not multiplied into the final physical score as independent priors.

### S4.3 A fitted residual can lie below thermal noise

For known $H$ and the true posterior mean $\mu(y) = \mathbb{E}[d \mid y]$, the residual $r = y - H(p + \mu(y))$ equals $\mathbb{E}[w \mid y]$. Total variance gives

$$N\sigma^2 = \mathbb{E}\,\|r\|^2 + \mathbb{E}\,\mathrm{tr}\{H\,\mathrm{Cov}(d \mid y)H^{\mathsf{H}}\}. \tag{S27}$$

Thus residual energy below $N\sigma^2$ is not automatically a noise-unit error. Unknown $H$, same-observation fitting, or approximate moments require additional terms or qualifications. The candidate scorer freezes its positive noise variance instead of estimating it from each candidate's fitted residual.

## S5 Separating physical-model error from evaluator error

Let $H$ and $G$ be fixed alternative channel operators, with a common message prior, codebook, pilot, and thermal-noise covariance. Put $\Delta H = H - G$, and use data mean $\mu_0$ and covariance $V_0$. Then

$$\mathcal{J}_H(G) = \mathbb{E}\,\|\Delta H(p + d)\|^2 = \|\Delta H(p + \mu_0)\|^2 + \mathrm{tr}(\Delta H V_0 \Delta H^{\mathsf{H}}). \tag{S28}$$

The fixed pilot contributes a directional term. For zero-mean isotropic data this becomes $\|\Delta Hp\|^2 + P_d\,\|\Delta H\|_F^2$, not generally a scalar multiple of Frobenius error alone.

For the joint message/observation laws $P_H$ and $P_G$, equal-covariance proper Gaussian divergence gives

$$D_{\mathrm{KL}}(P_H \| P_G) = \mathcal{J}_H(G)/\sigma^2. \tag{S29}$$

Averaging the conditional Gaussian divergence over messages proves the identity. Pinsker's inequality then gives, for the same fixed decision rule $\mathcal{A}$ under both propagation laws,

$$|R_H(\mathcal{A}) - R_G(\mathcal{A})| \le \min\{1, \sqrt{\mathcal{J}_H(G)/(2\sigma^2)}\}. \tag{S30}$$

For exact Bayes decisions under the respective models,

$$0 \le R_H(\delta_G^*) - R_H(\delta_H^*) \le \min\{1, \sqrt{2\mathcal{J}_H(G)/\sigma^2}\}. \tag{S31}$$

Insert and subtract the two risks under $G$; its Bayes-optimality makes the middle difference nonpositive and each remaining term is bounded by total variation.

These are physical-model-transfer bounds, not the evaluator's finite-list certificate. An estimate $G(y)$ obtained from the current observation cannot simply be substituted into (S29) as an exogenous Gaussian generative mean. In contrast, the main paper freezes the numerical arrays after observing $y$ and compares the resulting scores. Its mass bound remains algebraically valid even when the frozen model is imperfect. Neither argument gives a monotone ordering of FER as channel NMSE or $\mathcal{J}$ changes, and the general risk bounds can be vacuous.

# S6 Secondary hard fixed-work mode and numerical implementation

## S6.1 A separate operating contract

Suppose a terminal baseline fully evaluates the first $B_0$ words of a nested ordering, requiring $B_0 N$ row updates. Scan a longer list with hard incumbent pruning, stopping at that same row-update budget.

**Proposition S2** (Nested row-budget dominance)**.** *The returned best fully evaluated cost is no greater than the minimum full cost among the first $B_0$ words.*

*Proof.* Each of the first $B_0$ candidates costs at most $N$ updates. A stopped candidate cannot beat the incumbent. Thus all first-$B_0$ possibilities have either been evaluated or safely rejected before the budget expires. Additional completed words can only improve the incumbent; an unfinished additional word cannot remove the already available result. □

For a matched Bayesian score the selected word's conditional probability is no smaller, but a finite set of realized error indicators need not improve monotonically. Under model mismatch only the score ordering is guaranteed. Additional proposals and IDAFTs cost time, so this is not an equal-wall-clock result. Unvisited words do not have a completed soft certificate. It is intentionally not advertised as a second main receiver method.

The retained 192-draw controlled experiment reports:

| $B_0$ | Fixed-prefix failures | Expanded hard-search failures | Mean words assessed |
|---|---|---|---|
| 16 | 64 | 64 | 59.72 |
| 32 | 58 | 57 | 115.99 |
| 64 | 54 | 52 | 179.93 |

These small differences are exploratory; the exact mathematical result is the score dominance, not a population FER superiority claim.

## S6.2 Logarithmic implementation and failure handling

The reserve state should be stored as $\log Z_K$ and $\log U_P$. With $\log r = \log \epsilon - \log(1-\epsilon)$, the logarithm of available reserve is

$$\log(rZ_K - U_P) = \log r + \log Z_K + \log\left[1 - e^{\log U_P - \log r - \log Z_K}\right]. \tag{S32}$$

If the available reserve is nonpositive within numerical tolerance, do not prune. Use `log1p` and log-sum-exp to avoid cancellation/underflow. The output gain mixture uses only words processed through all $N$ rows.

The exact-arithmetic proof assumes a Hermitian positive-definite prior and exact Gaussian covariance updates. Rank-one downdates in finite precision can lose those properties. The supplied implementation checks predictive variances and uses a conservative comparison margin; it is not an interval-arithmetic proof. Stable square-root updates or a full factorization fallback should be charged explicitly. A numerical failure must return an error/uncertified status rather than a zero posterior contribution.

Ties use strict rejection and preserve the first minimizing candidate. A list must be duplicate-free unless multiplicities are deliberately combined with their prior weights. If all data-symbol combinations are not present, "MAP" always means MAP within that declared list. A nonuniform specified message prior can be added as an initial candidate cost $-\log \pi(d)$, leaving innovation increments nonnegative; the executed experiments use uniform priors.

# S7 Closest-method comparison and reproducibility contracts

## S7.1 Comparison at the complete-contribution level

| Method family | Search/likelihood object | Soft-information objective | Distinction relevant here |
|---|---|---|---|
| Blind/semiblind OFDM recursive search [5, 6] | Partial symbol sequences; channel fitting and partial-cost tests | Inspected construction targets an optimal hard/joint fitted solution | Complete AFDM codewords do not furnish independent symbol-tree observation prefixes; this paper integrates gains and bounds stopped weight sums |
| SO-OSD / GRAND / GCD / SCL [7] | Bitwise channel likelihood and structured/codebook models for unvisited words | Estimate confidence beyond an observed list | This paper uses exact coupled marginal prefixes but certifies only its declared expanded list; the coverage scopes are different |
| LC-SOSD [8] | Reliability-driven valid-codeword search; approximate APP-based stopping | Low-complexity soft output in a multiuser receiver | No general deterministic mass bound for the present coupled marginal score is imported from its bitwise approximation |
| Adaptive sphere clipping [9] | Known-channel MIMO max-log tree search | Adjust clipped LLR processing to a tracked error-rate target | Finite-list posterior sums and gain-mixture means, rather than max-log LLR clipping, are the present consumers |
| Dual-domain AFDM and data-aided SBL [10, 11] | Structured equalization or sparse channel inference | Detection/estimation accuracy versus cost | They are front-end/receiver alternatives, not an equivalent algorithm for certifying every incomplete marginal-list evaluation |

The table records inspected scope, not a claim that every possible extension of those methods is inferior. The numerical fixed-margin baseline in the main paper evaluates the same score; it is not relabeled as a reproduction of any one published soft sphere or OSD algorithm. The ordinary Transactions contribution is the complete coupled-model/fidelity/cost mechanism, while recursive conditioning, OSD and generic pruning remain acknowledged ingredients.

## S7.2 Retained data and new targeted evidence

The full controlled suite and timing measurements in the main paper originate from the selected V4 evidence package, not from the older foundational chapter. The original scripts and raw arrays are retained unchanged. The added tests are explicitly separate:

| Evidence | Contract | Status in this reconstruction |
|---|---|---|
| V4 controlled suite | 192 independent draws; list256; $N = 48, 96, 192$; two powers/noises/path counts | Retained, summarized from supplied raw/CSV records; uniform was frozen, adaptive originally post-synthesis |
| V4 balanced-path stress | 48 independent draws; $C_g = I/S$; mass and $10^{-3}$ gain tolerance | Retained; reveals little saving at high uncertainty |
| V4 timings | 15 interleaved repetitions on each fixture; proposal through gain output; front-end excluded | Retained; 34.74% and 21.90% are scoped fixture results |
| Estimated geometry | 32 new independent draws, namespace $(20260920, 7331, \ldots)$; three-atom OMP on a fixed coarse grid | Executed; all candidates scored under estimated, not actual, geometry; all five arms share observations/lists |
| Frozen I1 attachment | One completed $N = 1024$ full-fractional block, namespace $(20260920, 99173, 0)$ | Executed engineering test; all 24 returned integer atoms retained; no true geometry used by scorer |
| Independent numerical check | No import of candidate functions; explicit operators and observation covariance/augmented LS | Executed: 627 comparisons, maximum discrepancy $1.35 \times 10^{-11}$ |
| Foundation and clipped-metric check | Explicit positive-convention entries, exact finite alphabets, covariance identities, saved full-list minima | Executed: 72 comparisons, maximum discrepancy $9.10 \times 10^{-13}$ |

The initial two-block engineering invocation exceeded its call-time limit after saving the first completed block. No incomplete second-block results were admitted. A one-block invocation was then completed; its results are the engineering table used in the manuscript. The incomplete-attempt log remains in the package. This is an execution boundary, not a silently censored failed FER trial or evidence about average latency.

## S7.3 Frozen receiver identity and authorized inputs

The full-size attachment uses the supplied `TimingReceiver` I1 interface, recipe `B0-p2-s0.5`, legal `CodeFeedback` at checkpoint 1, and an outer-BP schedule of 12. The base model retains its frozen direct40 configuration. The checkpoint SHA256 is

`052ecfab8eff7780c49038d9bb4f16f92272ee22322fa05b6f65029cc3112a91.`

The $(2048, 1024)$ code-generator hash is

`bfd4247263adedc9a095b87603c2a7a4c2f2e8cb83f7e78bad721c5b06eaa6af.`

The receiver accepts exactly observation, scaled pilot, positive scalar thermal variance and constellation. The evaluator then receives its returned support and LLRs. True channel parameters and transmitted bits are stored solely for generation and offline error checks. A fixed $I/24$ gain prior is used instead of interpreting the receiver's estimated gains as independent prior evidence.

The scorer is attached terminally and its mean is not fed back into I1. This verifies a concrete observed-only interface. It does not demonstrate complete iterative integration or production performance. The single-frame OSD proposal is expensive (about 9.26 s), so scoring acceleration barely changes the approximately 12.93-s reference total. This is why the main paper reports both the positive moderate-dimensional scoring results and the weak full-size complete-cost result.

### S7.4 Timing fairness and statistical interpretation

The V4 timing baseline uses the same compiled recursion without pruning. Batch marginalization is a second same-likelihood implementation. The new estimated-geometry timings include pilot fitting, LMMSE, proposal, IDAFTs, scoring and bit-interval output. These new timings use three repeats per arm and a fixed arm order; they are descriptive engineering measurements, not randomized timing confirmation. Shared front-end terms are added equally to each arm. Static code/dictionary construction is excluded. The small estimated-geometry test also excludes its common final row-order sort; the frozen attachment excludes common support-to-regressor preparation and ordering. Reported new totals are therefore sums of measured stages, not independently measured stop-to-stop latencies. No unmeasured operation is credited as a speedup. The original 15-repeat V4 benchmark uses interleaved randomized ordering and remains the more carefully controlled component-cost evidence.

Frame-error fractions over mixed radio settings are descriptive aggregates. Zero disagreement between exact-reference and certified MAP is an implementation observation supporting a theorem, not an empirical claim of globally zero communication errors. Repeated tolerances or alternative policies on the same frames do not increase the number of independent channel draws. Numerical assertion counts are not independent research replications.

### S7.5 Reproduction commands

From the package root, run with a single BLAS/OpenMP thread:

```
export OPENBLAS_NUM_THREADS=1 OMP_NUM_THREADS=1 MKL_NUM_THREADS=1
python src/targeted_checks.py
python src/frozen_integration.py --frames 1
python src/independent_check.py
python src/clipped_maxlog.py
python src/independent_foundation.py
bash build.sh
```

The frozen attachment is CPU-specific in this package and rebuilds/verifies the supplied code. Initial code construction and JIT compilation are excluded from online timings but occur during reproduction. Current runtime outputs and hashes are saved separately; hardware-specific timings should not be expected to reproduce exactly.

### S7.6 Same-likelihood clipped max-log control

This control addresses a standard soft-search target without pretending that a new implementation reproduces a published sphere-tree receiver. Let $M_{i,v}$ be the smallest complete marginal cost so far among candidates with code bit $i$ equal to $v$, and let $\ell_*$ be the best complete cost. For a clipping level $L_{\max} = 5$, use the candidate-specific threshold

$$T_d = \max\left\{\ell_*,\ \max_i \min\{M_{i,c_i(d)}, \ell_* + L_{\max}\}\right\}. \tag{S33}$$

The first candidate is completed. Stop a later candidate when its observation-prefix cost exceeds $T_d$. Completed candidates update the counterhypothesis minima. For this finite-list adaptation, the returned clipped metrics equal the complete-list values

$$L_i^{\mathrm{ML}} = \mathrm{clip}\left(\min_{d:c_i(d)=1} \ell_N(d) - \min_{d:c_i(d)=0} \ell_N(d), -L_{\max}, L_{\max}\right). \tag{S34}$$

Here $c_i(d)$ denotes the binary code coordinate associated with complex candidate $d$. A stopped candidate cannot improve any minimum that remains relevant after clipping: its lower bound exceeds each

applicable capped minimum, and these caps can only decrease as further candidates are completed. The separate incumbent condition also preserves list MAP. Missing counterhypotheses yield saturated values; comparisons use the same tie convention as full evaluation.

All 32 estimated-geometry fixtures reproduce the exact clipped max-log metrics with maximum error $2.28 \times 10^{-13}$. The mean evaluated-row fraction is 0.7137, maximum discarded posterior mass is 0.140294, and three fixtures exceed the main paper's 1% mass limit. The maximum posterior channel-mean change is 0.004409. This tests the distinction between clipped max-log fidelity and probability-sum fidelity; it is not evidence that clipping fails its own stated target. The control was added after the initial 32-fixture evaluation. No additional independent radio draws or dedicated wall-clock benchmark are claimed for it.

## S8 Claim map and remaining confirmation

| Paper claim | Mathematical basis | Evidence artifact |
|---|---|---|
| Exact fractional path rows with one convention | Main model and (S3) | `independent_check.py`; original physical reference |
| Monotone complete marginal evaluation | Main Theorem 1 | V4 `independent_verification.json`; new dense/augmented checks |
| Finite-list MAP and controlled mass | Main Theorem 2 | V4 and new full/partial costs; `estimated_geometry.npz` |
| Bit-probability/LLR intervals | Main Corollary 1 | All new saved bit intervals checked against complete-list probabilities |
| Bounded posterior channel-mean change | Main Theorem 3 | V4 balanced stress; new geometry/interface checks |
| Reduced computation in intended regime | Cost formula and measured implementation | V4 `timing_raw.csv`; new stage times |
| Full-size observed-only attachment | Specified frozen adapter, not a new estimator | `frozen_integration.*` |
| Complete-I1 population speedup or FER advantage | Not established | No such claim or table in this paper |

A ready-to-run larger independent-confirmation driver remains in the unchanged V4 package. It is not executed at its planned sample size in this reconstruction. The immediately relevant production comparison is narrower than another broad method search: use a frozen front end and geometry-acquisition interface, compare complete and certified evaluation of the same proposed lists, measure all material stages on the intended hardware, and then assess whether the cheaper evaluator enables a useful larger-list operating point. If the proposal dominates or the posterior is diffuse, the evaluator may have little system-level benefit despite a correct guarantee.

This supplement deliberately excludes the old integer-domain candidate receiver, support-exchange formulas, inactive self-generated penalties, scorer/learning chronology, and rejected pilot/coreset designs. Their historical records are preserved only in source archives. None is a second proposed method or a contribution claimed by the present paper.

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